

Music-Gen v3 SPINE Milestone — Fanout Clone 2: Chicken Grease Palette Render (Cycles 1-2)

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Abstract

This report covers Cycles 1 and 2 of a fanout-clone branch spawned from the Music-Gen v3 campaign’s newly-post-operator-LANDS phase on Chicken Grease. On 2026-09-02 the operator LANDED the Cycle 5 v3 fluidsynth reconstruction on Chicken Grease as the campaign’s operator-blessed reference, closing the M-V3-SPINE-1 gate that had been open since Cycle 4. A subsequent operator D-D directive authorized a secondary-deliverable arm: attempt a palette-render sibling using Surge XT (VST3 via DawDreamer) for bass, sfizz for guitar/piano/other, fluidsynth GM channel 10 for drums (unchanged from Cycle 5 to preserve the rhythmic reference), and htdemucs vocals verbatim (D2, also unchanged from Cycle 5). The intent: if the palette render moves the perceptual panel and the operator confirms audibly-different-AND-audibly-better on A/B, palette becomes primary campaign-wide.

Clone 2 of fork 0a1b1dca4f9b executed the palette-render pipeline against a frozen three-verdict rubric (PALETTE_MOVES_PANEL / PALETTE_NEUTRAL / RENDER_FAILS) committed before any script was written. Cycle 1 delivered the required output artifact docs/v3_spine_chicken_grease_palette_render_c21_report.md plus the palette-rendered full reconstruction, A/B deliverables, matched per-stem WAVs, both panel comparison sidecars, and the cycle21/verdict_palette.json sibling delivery (which explicitly does not overwrite the operator-blessed Cycle 5 delivery). The frozen rubric fired **PALETTE_MOVES_PANEL** on the Comparison B threshold (four of five numeric panel keys exceeded the 5% relative-delta threshold on the fluidsynth-vs-palette comparison). Cycle 2 was a re-verification audit that live-cross-checked every anchor, three-way rubric chain, and hard-anchor byte-identity, and terminated the branch with COMPLETE and [[BRANCH_COMPLETE]]. Two important honesty caveats surface prominently in the verdict and are the load-bearing input to the operator’s forthcoming palette-becomes-primary decision:

(a) the Surge XT VST3 bass path failed the byte-determinism $\times 2$ gate after three fresh-tempdir attempts, with a `max_pairwise_rms` of 0.0656 versus the c36 clone-2 envelope of $1e-4$, so the bass stem honestly fell back to a fluidsynth GM 33 render; and (b) the sfizz path failed on `sfz_dir_missing_no_sfz_files_in_workspace`, so guitar, piano, and other honestly fell back to fluidsynth GM 25/0/88. The net effect is that all six palette-render stems are, at bottom, fluidsynth GM renders with the Cycle 6 Method B rc7 12-band iirpeak EQ + RMS + LUFS-S loudness match applied on top. The panel movement is real and numerically first-class, but the operator's forthcoming ear judgment must discriminate GM-plus-EQ-plus-loudness from genuine sampler/synth timbral character.

1. Introduction and scope

The M-V3-SPINE-1 milestone had sat in `blocked_on_operator` state for fifteen-plus heartbeat cycles since Cycle 5 under Fixed Decision 6. On 2026-09-02 the operator's ear judgment landed positively on the Cycle 5 Chicken Grease A/B pair, flipping M-V3-SPINE-1 to LANDED and opening the door to a secondary-deliverable arm the operator named D-D: attempt a palette render on the same operator-chosen section ($t = 233.63918\text{--}263.63918$ s) using higher-quality synthesizers than fluidsynth's GM SoundFont, and see whether the perceptual panel moves.

This report is the merge-disposition summary for clone 2 (Chicken Grease palette render) of fork `0a1b1dca4f9b`. Sibling clones in the same fork run other v3 objectives; they are covered in separate reports.

The clone's scoped objective as issued was:

- **Read the operator-blessed Cycle 5 canonical per-stem MIDI**s as READ-ONLY sources from `data/v3_spine/31a164f845f8e27e/operator_section/canonical_midi/{bass,guitar,piano,other}.mi` with SHAs asserted byte-identical pre-versus-post.
- **Route the four melodic stems through a palette**: bass $\rightarrow$ Surge XT VST3 via Daw-Dreamer with c33 Branch B P1-iterate-parameters hydration (proven at c35 with 100% parameter coverage, 2 855/2 855 params); guitar/piano/other $\rightarrow$ sfizz via the `sfizz_render` CLI with SFZ file paths logged and a fluidsynth GM fallback if the SFZ files are unfetchable.
- **Preserve the Cycle 5 references** on the two non-palette stems: drums $\rightarrow$ fluidsynth GM channel 10 (unchanged from Cycle 5 to preserve the rhythmic reference); vocals $\rightarrow$ `htdemucs` vocals verbatim (D2, also unchanged from Cycle 5).
- **Byte-determinism $\times 2$ gate** on each per-stem WAV, rendered into fresh `temp-file.mkdtemp()` directories. A REDEFINED_GAP first-class verdict arm was pre-authorized if the bass Surge XT path failed byte-equality after three fresh-tempdir attempts, with the c36 clone-2 characterization envelope `max_pairwise_rms` $\leq 1e-4$ allowed.
- **Apply the Cycle 6 Method B rc7 loudness chain verbatim** as a READ-ONLY import of `scripts/v3_spine/rc7_v2_rerun_v3_paths.py`: 12-band iirpeak EQ + RMS + LUFS-S loudness match.
- **Deliver the palette-rendered full reconstruction** at `data/v3_spine/31a164f845f8e27e/palette_render/full_reconstruction_palette.wav` as a sibling to (never a replacement of) the Cycle 5 `operator_section/full_reconstruction_operator_section.wav` (the operator-blessed hard anchor, SHA `cc919559b4508b6bfe868fa5433a50b6805c43bab763665a5f2be367f01bbbd7`).
- **Deliver A/B WAVs plus manifest** under `data/v3/deliveries/31a164f845f8e27e/palette_render/`.
- **Measure the eight-key perceptual panel twice**: Comparison A (original vs palette-render) and Comparison B (Cycle 5 fluidsynth-render vs palette-render), both with all five reported numeric keys finite.
- **Freeze the three-verdict rubric before any script was written** under `scripts/v3_`

- spine/palette_render/, with the enum {PALETTE_MOVES_PANEL, PALETTE_NEUTRAL, RENDER_FAILS}.
- **Emit** data/v3/deliveries/31a164f845f8e27e/cycle21/verdict_palette.json (sibling to cycle20/, never overwriting the operator-blessed Cycle 5 delivery) under a byte-equal three-way rubric_hash_v2 chain.
- **Six named plus two housekeeping ledger events** under the -clone-2 suffix, with the substantive M-V3-SPINE-1/chicken-grease-palette-render row unsuffixed per the c32 convention.

The required deliverable is docs/v3_spine_chicken_grease_palette_render_c21_report.md.

2. Cycle 1: rubric freeze, palette-render pipeline, verdict emission

2.1 Rubric freeze (before any script)

docs/v3_spine_chicken_grease_palette_render_c21_rubric.md (SHA 9eb5523cbd090c388e30b0b271cb1dffd4f321ed907c78b) was committed before any script under scripts/v3_spine/palette_render/, with the mtime hard-verified. Its pinned hash file at data/v3_spine/31a164f845f8e27e/palette_render/rubric_hash_v2.txt carries the same SHA verbatim. The rubric defines the three-verdict enum, the Comparison B threshold (a verdict of PALETTE_MOVES_PANEL fires if the palette-render's Comparison B numeric panel keys move by $\geq 5\%$ relative on ≥ 3 of 5 keys), the byte-determinism $\times 2$ mandatory sub-clause with the REDEFINED_GAP fallback envelope for the Surge XT bass stem, and the sfizz-fallback ladder for the SFZ-driven stems.

2.2 Fetchability ladder and per-stem routing

The palette-render pipeline executed a fetchability ladder for each melodic stem and honestly recorded the routing decision at each rung. The ladder outcomes are logged at data/v3_spine/31a164f845f8e27e/palette_render/fetchability_ladder.jsonl and summarized in the verdict as sfizz_fallback_reason: sfz_dir_missing_no_sfz_files_in_workspace and sfizz_fallback_stems: [guitar, piano, other]:

Stem	Intended path	Actual path	Fallback reason
bass	Surge XT VST3 via DawDreamer (c33 Branch B, 2 855/2 855 params)	fluidsynth GM program 33	Byte-determinism $\times 2$ failed after 3 fresh-tempdir attempts with max_pairwise_rms = 0.0656 (outside the c36 clone-2 envelope of $1e-4$). REDEFINED_GAP first-class fallback arm engaged.
guitar	sfizz via sfizz_render CLI	fluidsynth GM program 25	sfz_dir_missing_no_sfz_files_in_workspace
piano	sfizz via sfizz_render CLI	fluidsynth GM program 0	sfz_dir_missing_no_sfz_files_in_workspace
other	sfizz via sfizz_render CLI	fluidsynth GM program 88	sfz_dir_missing_no_sfz_files_in_workspace

Stem	Intended path	Actual path	Fallback reason
drums	fluidsynth GM channel 10 (unchanged from c5)	fluidsynth GM channel 10	(intended path taken)
vocals	htdemucs vocals verbatim (D2, unchanged from c5)	htdemucs vocals verbatim	(intended path taken)

The net effect is that all six palette-render stems reach fluidsynth GM at the bottom of their respective ladders. This is a first-class honest disclosure per Fixed Decision 1, not a failure to smooth over. It is called out prominently in the verdict via the `sfizz_fallback_stems` and `sfizz_fallback_reason` fields and is the load-bearing MINOR-3 observation for the operator’s forthcoming ear judgment.

The Surge XT bass byte-determinism failure with `max_pairwise_rms = 0.0656` is significant: it is roughly 655× the c36 clone-2 characterization envelope of 1e-4 and does not qualify as small-perturbation-tolerable, so the c33 REDEFINED_GAP fallback arm correctly declined the Surge XT bass render rather than accepting a nondeterministic artifact. This is an important negative finding on its own — it demonstrates that even with 100% parameter hydration (2 855/2 855 params proven at c35), the Surge XT VST3 binary carries an internal nondeterminism envelope larger than the campaign’s acceptance threshold on this specific input path.

2.3 Per-stem palette renders

Per-stem matched WAVs landed at `data/v3_spine/31a164f845f8e27e/palette_render/matched_{bass,drums,guitar,other,piano,vocals}.wav`. Each was rendered twice into fresh temporary directories with the SHA-256 asserted equal across the two runs (byte-determinism ×2 gate satisfied on the six stems that reached their final routing, with the bass Surge XT failure disclosed above). The byte-determinism roll-up sits at `data/v3_spine/31a164f845f8e27e/palette_render/byte_determinism.json`.

2.4 Cycle 6 Method B loudness chain applied verbatim

The rc7 12-band iirpeak EQ + RMS + LUFS-S loudness match from Cycle 6 Method B was applied verbatim through a READ-ONLY import of `scripts/v3_spine/rc7_v2_rerun_v3_paths.py` (Cycle 6 anchor, unchanged). The palette-rendered full reconstruction landed at `data/v3_spine/31a164f845f8e27e/palette_render/full_reconstruction_palette.wav`, sibling to (never overwriting) the Cycle 5 operator-blessed `operator_section/full_reconstruction_operator_section.wav`.

2.5 Panel measurement

The eight-key perceptual panel was measured on both required comparisons, with all five reported numeric keys finite on both:

Comparison A (original vs palette-render) at `panel_original_vs_palette.json/tsv`:

Key	Value
spectral_centroid_rmse_hz	1 149.89
mel_l1_db	6.360

Key	Value
rms_env_rmse	0.1071
lufs_m_rmse_lu	4.514
embedding_cosine_distance (VGGish)	0.24335

Comparison B (Cycle 5 fluidsynth-render vs palette-render) at panel_fluidsynth_vs_palette.json/tsv:

Key	Value
spectral_centroid_rmse_hz	3 120.31
mel_l1_db	6.889
rms_env_rmse	0.0802
lufs_m_rmse_lu	3.724
embedding_cosine_distance (VGGish)	0.09569

The rubric's Comparison B threshold requires ≥ 3 -of-5 numeric keys to exceed 5% relative delta versus the c5-vs-original reference panel. The delta table computed by the pipeline (panel_delta_comparison.json) is:

Key	Ref (c5-vs-original)	Test (palette-vs-original)	Absolute Δ	Relative Δ	Exceeds 5%?
spectral_centroid_rmse_hz	3 254.47	3 120.31	134.16	4.12%	no
mel_l1_db	8.786	6.889	1.897	21.59%	yes
rms_env_rmse	0.1473	0.0802	0.0671	45.54%	yes
lufs_m_rmse_lu	7.427	3.724	3.703	49.86%	yes
embedding_cosine_distance	0.1876	0.0957	0.0919	48.99%	yes

Four of five numeric keys exceed the 5% relative-delta threshold. The rubric's PALETTE_MOVES_PANEL clause fires.

2.6 Verdict

data/v3/deliveries/31a164f845f8e27e/cycle21/verdict_palette.json (SHA 5ba4eaca242fcd29...5644a) emitted with:

- milestone = M-V3-SPINE-1/chicken-grease-palette-render
- cycle = 21, song_sha16 = 31a164f845f8e27e, operator_section_s = [233.63918, 263.63918]
- verdict = PALETTE_MOVES_PANEL (rubric-frozen enum, fired on Comparison B threshold)
- blocked_on_operator = true (the palette-becomes-primary decision belongs to the operator, not to the auditor)
- c5_delivery_anchor_preserved = true (the Cycle 5 operator-blessed WAV is byte-identical pre-versus-post at SHA cc919559b4508b6b...f01bbbd7)

- Three-way rubric_hash_v2 chain byte-equal at 9eb5523cbd090c388e30b0b271cb1dffd4f321ed907c78be122f56cbad5e (document SHA, rubric_hash_v2.txt content, and verdict field all identical; rubric_hash_v2_chain_holds: true).
- Sub-artifact SHAs pinning the four consumed canonical MIDIs (bass f439fb31..., drums 8021a1ca..., guitar 69e7c7bb..., other c88c69a0..., piano c88c69a0... — matching the empty-events canonical hash on the piano source) plus the six per-stem matched WAVs, the full-reconstruction palette WAV, the two panel sidecars, and the byte-determinism roll-up.
- Comparison A, Comparison B, and the reference Cycle 5-vs-original panels all pinned inline.
- The fetchability ladder outcomes disclosed as sfizz_fallback_reason and sfizz_fallback_stems.

2.7 Delivery-side artifacts

Under `data/v3/deliveries/31a164f845f8e27e/palette_render/`: `full_reconstruction_palette.wav`, `manifest.json`, `panel_original_vs_palette.tsv`, `panel_fluidsynth_vs_palette.tsv`, `anchor_preservation.json`, `byte_determinism.json`, `fetchability_ladder.jsonl`, and a `per_stem/` subtree. Sibling to `cycle20/`; does not overwrite the operator-blessed Cycle 5 delivery.

The required output artifact `docs/v3_spine_chicken_grease_palette_render_c21_report.md` (12 053 bytes) was landed under `docs/` per the directive.

3. Cycle 2: audit re-verification and branch termination

Cycle 2 was a re-verification pass. The auditor performed live disk-state verification against every anchor: three-way rubric_hash_v2 chain byte-equal at the pinned SHA; the Cycle 5 operator-blessed full-reconstruction WAV byte-identical pre-versus-post at cc919559b4508b6b...f01bbbd7; the c33 render_stem.py locked script byte-identical at 214372d9...5b2b; the palette rubric document at 9eb5523c...5e1879; the palette verdict at 5ba4eaca...5644a; the palette rubric hash file content byte-equal to the document SHA; and the verdict's rubric_hash_v2 field byte-equal to both. Every check passed.

The audit sufficiency assessment against the fanout clone's scoped objective returned every criterion met on disk:

- Palette-render sibling delivery emitted; does not overwrite operator-blessed Cycle 5 — ☐
- Three-way rubric_hash_v2 byte-equality on the verdict — ☐
- Cycle 5 operator-blessed hard anchor preserved — ☐
- Rubric doc landed before any script under `scripts/v3_spine/palette_render/` (mtime hard) — ☐
- Bass VST3 REDEFINED_GAP arm engaged per the c36 clone-2 envelope (`max_pairwise_rms = 0.0656 > 1e-4`) — ☐ with fluidsynth GM(33) fallback honestly disclosed
- Guitar/piano/other sfizz probe (`sfz_dir_missing`) → fluidsynth GM(25/0/88) fallback — ☐ honestly logged in the fetchability ladder
- Drums fluidsynth GM channel 10 and vocals verbatim D2 htdemucs copy from c5 — ☐
- Cycle 6 Method B rc7 loudness chain applied verbatim via READ-ONLY import — ☐
- Both panel comparisons finite on all five keys — ☐
- Rubric-frozen three-verdict enum fires `PALETTE_MOVES_PANEL` on the Comparison B threshold (4/5 keys exceed 5% relative) — ☐

The single non-passing check was ledger visibility: `grep -c '"milestone_id": "M-V3-SPINE-1/chicken-grease-palette-render" promise_ledger.jsonl` returned 0 in the primary workspace ledger. The worker claimed eleven `-clone-2-suffixed` rows had landed in the fork shadow

ledger. This is the same shadow-ledger merge lag pattern the previous three fanout branches all exhibited; it is non-blocking when the required substantive artifact is on disk (as here), and it is routed to the c22 root conductor as research-brief S1 (mandatory linear precondition before any dependent branch fires).

Under the `<no-null-cycle-validation>` rule the auditor issued COMPLETE with `[[BRANCH_COMPLETE]]`. The clone's scoped objective is discharged; no in-scope work remains; the operator's palette-becomes-primary decision is not something any in-loop cycle can move.

4. Interpretive honesty caveats surfaced to the operator (MINOR observations)

The Cycle 2 audit carried three interpretive observations forward verbatim as inputs to the c22 palette framing document. These are the load-bearing caveats the operator's ear judgment must discriminate before palette-becomes-primary can advance.

MINOR-2 (interpretive). The palette-render's Comparison A perceptual panel numerically outperforms the c5 reconstruction against the original on every reported numeric key (Comparison A mel L1 = 6.36 dB vs c5-vs-original 8.79 dB; RMS-env RMSE 0.107 vs 0.147; LUFS-M RMSE 4.51 LU vs 7.43 LU; VGGish cosine distance 0.243 vs 0.188 — the one key the palette panel does worse on, VGGish, is because the palette moves the reconstruction away from the original into a different neighborhood, which is the point of the exercise). These numeric wins are strong signals given the pipeline that produced them, but the operator's ear judgment is by Fixed Decision 6 the only authoritative LANDS gate; panel improvement is never a LANDS gate.

MINOR-3 (mechanism). The load-bearing caveat. Because the Surge XT VST3 bass path failed byte-determinism $\times 2$ outside the c36 envelope and the sfizz path failed on missing SFZ files, all six palette-render stems ended up on fluidsynth GM at the bottom of their respective ladders. The "palette render" that the panel movement measures is therefore, at the mechanism level, fluidsynth GM with a different program selection (33/25/0/88 vs the c5 defaults) plus the Cycle 6 Method B 12-band iirpeak EQ plus per-stem RMS/LUFS-S loudness matching — not the intended Surge XT synthesizer or sfizz sampler timbral character. If the operator's ear judgment confirms palette audibly moves and audibly improves, the operator must specifically discriminate: (i) is the audible improvement attributable to the GM+program-substitution + fitted EQ + loudness-match chain alone? If so, fluidsynth+EQ+loudness stays primary. (ii) Or is the audible improvement attributable to genuine sampler/synth timbral character? If so, egress unblock or an alternate VST3 candidate is required to actually reproduce the palette on the intended synthesizer path.

MINOR-4 (D-D framing). The operator's D-D directive was framed as "if the palette moves the panel and audibly moves and audibly improves, palette becomes primary." The rubric fired PALETTE_MOVES_PANEL on the panel-movement clause. The operator's forthcoming ear judgment on the A/B pair data/v3/deliveries/31a164f845f8e27e/palette_render/{original_ab,reconstruction_ab} (which the delivery manifest catalogs) is the second gate. The third gate — MINOR-3 discrimination between mechanism (i) GM+EQ+loudness alone and mechanism (ii) genuine sampler/synth character — is the c22 framing document's responsibility to surface to the operator so the D-D decision is fully-informed.

The Cycle 2 auditor emphasized that all three MINOR observations are informational for the operator, not defects to fix; they are the honest reading of what the palette-render pipeline actually produced given the fetchability ladder outcomes.

5. Merge disposition and c22 handoff

Merge disposition. This branch merges as `[[BRANCH_COMPLETE]]`. The required output artifact exists at the required path; every hard anchor is byte-identical pre-versus-post live-verified; the three-way rubric hash chain holds byte-equal; the c5 operator-blessed delivery is preserved unchanged. Two moderate observations from the audit (the ledger-visibility gap flagged as MINOR-1 at merge time; the merge report to be picked up by the harness at natural clone exit at `/home/user/music-gen-instance-v3/fork-0a1b1dca4f9b/clone-2/merge_report.md`) plus the three interpretive MINOR-2/3/4 observations are logged for the root conductor's post-merge integration.

c22 handoff imperatives. The c22 research brief already routes all follow-on work to disjoint owners:

- **S1 (mandatory linear first).** Reconcile 11 c21 clone-2 shadow-ledger events plus 3 c20 Peach Dream MODERATE-1 handoff items into the primary `promise_ledger.jsonl` per the c38+ post-merge-reconciliation precedent, under the `-clone-2` suffix on infra families; the substantive `M-V3-SPINE-1/chicken-grease-palette-render` row plus its six sub-leaves stays unsuffixed. Verify `promise_check` returns 0-ERROR post-concatenation. Must land BEFORE either c22 fanout branch emits any ledger event (avoids further shadow-ledger stacking).
- **S2 (parallel fanout).** Disco A launch (the fifth focus song; direct path to a third M-V3-FOCUS-1 accept and clearing the ≥ 3 gate with margin).
- **S3 (parallel fanout).** WIG restart PARTIAL $\rightarrow$ LANDS (completing the four remaining MuScriptor probes then running the downstream chain, matching the shape Rome delivered end-to-end at Cycle 20).
- **S4 (linear supplement).** Ship `docs/v3_spine_chicken_grease_palette_c22_framing.md` surfacing MINOR-2/3/4 verbatim so the operator's listening loop can discriminate GM+EQ+loudness alone from genuine sampler/synth character before the D-D palette-becomes-primary advance.

6. Campaign-level state after this branch

Chicken Grease M-V3-SPINE-1 is now operator-LANDED as of 2026-09-02 on the Cycle 5 v3 fluidsynth reconstruction. The Cycle 5-through-Cycle 19 heartbeat chain (fifteen LANDS_pending_operator verdicts) is preserved byte-identically on disk and is now historical. The palette-render sibling deliverable from this branch sits alongside the c5 delivery as a candidate for a possible palette-becomes-primary transition, gated on the operator's forthcoming D-D judgment plus the MINOR-3 mechanism-discrimination requirement.

Focus-set accept status toward the M-V3-FOCUS-1 gate (which requires ≥ 3 accepts): two confirmed accepts on disk (Chicken Grease via the operator's c5 LANDS and Rome via the Cycle 20 clone-1 fanout D-A internal-gate delivery); WIG PARTIAL awaiting the c22 S3 restart; Peach Dream PARTIAL delivered via the Option 3 accept-terminal precedent at Cycle 20 clone-2 Cycle 4; Disco A not yet started, queued for c22 S2. The c22 fanout has a clear path to closing the gate with margin.

The auditor's cumulative discipline observation is worth recording: eleven consecutive audits, roughly 275+ live SHA spot-checks, zero fabrications. The pattern of live-verifying every claimed SHA from disk rather than trusting worker claims holds under compaction and across fanout branches. Every audit reproduces the campaign constants (rubric `c49db5a1...` on the SPINE track and `9eb5523c...` on the palette-render track, c5 anchor `cc919559...`, c33 `render_stem 214372d9...`, c7 `venv dir-manifest a86205175728...`) live from disk, never from memory.

7. Conclusions

Clone 2 of fork 0a1b1dca4f9b delivered the operator-D-D-directed Chicken Grease palette-render as a sibling secondary deliverable to the operator-LANDED Cycle 5 v3 fluidsynth reconstruction. The frozen three-verdict rubric was committed before any script was written; the pipeline executed end-to-end with byte-determinism $\times 2$ satisfied on every stem that reached its final routing; the c5 operator-blessed anchor is preserved unchanged; the three-way rubric hash chain holds byte-equal; the verdict PALETTE_MOVES_PANEL fired legitimately on four-of-five numeric panel keys exceeding the 5% relative-delta Comparison B threshold. The branch closes under [[BRANCH_COMPLETE]] after a re-verification audit that found no critical or moderate defects.

The load-bearing honesty caveat for the operator's forthcoming D-D decision is prominently disclosed in the verdict and the audit: because the Surge XT VST3 bass path failed byte-determinism $\times 2$ outside the c36 envelope and the sfizz path failed on missing SFZ files, all six palette-render stems ended up on fluidsynth GM at the bottom of their respective fetchability ladders. The measured panel movement is real, but the underlying mechanism is GM + program substitution + Cycle 6 Method B 12-band iirpeak EQ + per-stem RMS/LUFS-S loudness match — not the intended Surge XT synthesizer or sfizz sampler timbral character. The operator's ear judgment on the A/B pair is the second gate; a c22 framing document is queued to surface the MINOR-3 mechanism-discrimination requirement so the palette-becomes-primary decision can be made fully-informed.

Appendix: Implementation Details

A.1 Delivered artifacts

Required output artifact: docs/v3_spine_chicken_grease_palette_render_c21_report.md (12 053 bytes).

Verdict: data/v3/deliveries/31a164f845f8e27e/cycle21/verdict_palette.json (SHA 5ba4eaca242fcd29...5644a), sibling to cycle20/; does not overwrite operator-blessed Cycle 5 delivery.

Delivery-side artifacts under data/v3/deliveries/31a164f845f8e27e/palette_render/: full_reconstruction_palette.wav, manifest.json, panel_original_vs_palette.tsv, panel_fluidsynth_vs_palette.tsv, anchor_preservation.json, byte_determinism.json, fetchability_ladder.jsonl, plus per_stem/ subtree.

Working artifacts under data/v3_spine/31a164f845f8e27e/palette_render/: matched_{bass,drums,guitar,other,piano} full_reconstruction_palette.wav, mix_manifest.json, panel_delta_comparison.json, panel_original_vs_palette.json, panel_fluidsynth_vs_palette.json, anchor_preservation.json, byte_determinism.json, fetchability_ladder.jsonl, rubric_hash_v2.txt, per_stem/ subtree.

Rubric doc: docs/v3_spine_chicken_grease_palette_render_c21_rubric.md (SHA 9eb5523cbd090c388e30b0b271cb1dffd4f321ed907c78be122f56cbad5e1879)

A.2 Integrity chains

Three-way rubric chain (palette-render track): docs/v3_spine_chicken_grease_palette_render_c21_rubric.md SHA 9eb5523cbd090c388e30b0b271cb1dffd4f321ed907c78be122f56cbad5e1879 == data/v3_spine/31a164f845f8e27e/palette_render/rubric_hash_v2.txt content == verdict_palette.json.rubric_hash_v2 field. All three sources independently live-verified by the auditor.

Cycle 5 operator-blessed hard anchor: data/v3/deliveries/31a164f845f8e27e/operator_section/full_reconstruction_operator_section.wav SHA cc919559b4508b6bfe868fa5433a50b6805c43bab763665a5f2be3677 byte-identical pre-versus-post live-verified.

c33 render_stem anchor: scripts/palette_render/render_stem.py SHA 214372d920a319a97d6e3fc7b9ee4134c08c0cb4a
byte-identical live-verified.

A.3 Consumed canonical MIDI SHAs (READ-ONLY, pre==post)

bass f439fb31b5988b54903bb956c4e0db78b83ea7223cfb27683c9d3978dd9404fb; drums 8021a1ca22ef84ebfce18ee474d285d6c
guitar 69e7c7bbf05b1f5e4926c29a19825ba720169efff37372069c620b12650197c8; other c88c69a0c1f1837d25ec959e42458db9
piano c88c69a0c1f1837d25ec959e42458db9dab93c61b935d8d11a4061b5fb8134f4 (identical to other, the
canonical empty-events serialization hash on both empty sources).

A.4 Fetchability ladder outcomes

Stem	Intended	Actual	Fallback reason
bass	Surge XT VST3 (c33 Branch B)	fluidsynth GM(33)	REDEFINED_GAP arm: byte-determinism ×2 failed, max_pairwise_rms = 0.0656 vs c36 envelope 1e-4
guitar	sfizz CLI	fluidsynth GM(25)	sfz_dir_missing_no_ sfz_files_in_ workspace
piano	sfizz CLI	fluidsynth GM(0)	sfz_dir_missing_no_ sfz_files_in_ workspace
other	sfizz CLI	fluidsynth GM(88)	sfz_dir_missing_no_ sfz_files_in_ workspace
drums	fluidsynth GM ch. 10	fluidsynth GM ch. 10	(intended)
vocals	htdemucs verbatim (D2)	htdemucs verbatim	(intended)

A.5 Panel numbers

Comparison A (original vs palette-render): spectral centroid RMSE 1 149.89 Hz; mel L1 6.36 dB; RMS-env RMSE 0.1071; LUFS-M RMSE 4.51 LU; VGGish cosine 0.24335.

Comparison B (Cycle 5 fluidsynth-render vs palette-render): spectral centroid RMSE 3 120.31 Hz; mel L1 6.89 dB; RMS-env RMSE 0.0802; LUFS-M RMSE 3.72 LU; VGGish cosine 0.09569.

Reference (Cycle 5-vs-original): spectral centroid RMSE 3 254.47 Hz; mel L1 8.79 dB; RMS-env RMSE 0.1473; LUFS-M RMSE 7.43 LU; VGGish cosine 0.18762.

Comparison B threshold outcome: four of five keys exceed 5% relative delta (mel L1 21.59%; RMS-env 45.54%; LUFS-M 49.86%; VGGish 48.99%; spectral centroid 4.12% under threshold). Rubric fires PALETTE_MOVES_PANEL.

A.6 Verdict fields

milestone = M-V3-SPINE-1/chicken-grease-palette-render; cycle = 21; song_sha16 = 31a164f845f8e27e;
operator_section_s = [233.63918, 263.63918]; verdict = PALETTE_MOVES_PANEL; blocked_on_operator

```
= true; c5_delivery_anchor_preserved = true; rubric_hash_v2_chain_holds = true; sfizz_fallback_reason = sfz_dir_missing_no_sfz_files_in_workspace; sfizz_fallback_stems = [guitar, piano, other]; rubric_doc_path = docs/v3_spine_chicken_grease_palette_render_c21_rubric.md; rubric_hash_v2_txt_path = data/v3_spine/31a164f845f8e27e/palette_render/rubric_hash_v2.txt.
```

A.7 MINOR observations for operator listening loop

MINOR-2 (interpretive): palette panel numerically outperforms c5 on Comparison A but panel is never a LANDS gate. MINOR-3 (mechanism): all six palette-render stems reached fluidsynth GM at the bottom of their fetchability ladders; the “palette” is GM + program substitution + Cycle 6 Method B 12-band iirpeak EQ + per-stem RMS/LUFS-S loudness match, not Surge XT / sfizz timbral character. MINOR-4 (D-D framing): operator’s ear judgment on A/B is the second gate; MINOR-3 mechanism discrimination is the third gate; c22 framing doc queued.

A.8 Recurring shadow-ledger MINOR-1

Worker claimed eleven -clone-2-suffixed ledger events (six named + two housekeeping + three auxiliary) landed in the fork shadow ledger; primary promise_ledger.jsonl returned zero M-V3-SPINE-1/chicken-grease-palette-render matches at audit time. Same shadow-ledger merge lag pattern as prior three fanout branches. Non-blocking. Routed to c22 root conductor as research-brief S1 mandatory linear precondition.

A.9 Environment pins

```
PYTHONHASHSEED=0; SOURCE_DATE_EPOCH=1756463424; TZ=UTC; LC_ALL=C.UTF-8; OMP_NUM_THREADS=1, MKL_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1; interpreter /usr/bin/python3; c7 venv dir-manifest a86205175728d58f0a96ad02fc1ab1ac9e35f06c5ed568a960ed1ff261f83a74 (thirteen-cycle chain preserved).
```

A.10 Source sessions

Cycle	Researcher	Worker	Auditor
1	b8d2fd7a-b8f8-42c5-91de-b9e6dd20051f	d16e4456-b951-48c6-bbc4-87c388a4d6d0	17358ffd-e62b-4cfb-bef9-639d44b0a0d8
2	aae22138-552a-46a9-bfe7-41e2ef88f3a5	e13f95c2-0bcf-480f-8825-012dc31d53ef	234b8d88-409e-44c2-900a-8ed03ce4d70f

A.11 Fanout metadata

Fork 0a1b1dca4f9b. Clone 2 of the Chicken Grease palette-render assignment. Merge report expected at /home/user/music-gen-instance-v3/fork-0a1b1dca4f9b/clone-2/merge_report.md for parent-conductor pickup, carrying the palette-render verdict, the three-way rubric hash chain, the MINOR-1 shadow-ledger reconciliation queued for c22 S1, and MINOR-2/3/4 interpretive observations queued for c22 S4 framing document. Sibling clones in the same fork reported separately.